

Amritpal Singh Kaur

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English · Spanish · Catalan · Punjabi (Native) | Hindi (Professional)

PROFESSIONAL SUMMARY

MSc Advanced Computer Science candidate (Queen Mary University of London, 2026) with 3+ years of enterprise IT experience in regulated environments including Bank of America and Amadeus. Proficient in Python, JavaScript, TypeScript, AWS, Docker, and CI/CD pipelines. Strong background in security-hardened deployments, AI-integrated full-stack development, and formal software verification. Seeking 2026 graduate roles in Software Engineering, Cloud/DevOps, or Cybersecurity.

EDUCATION

MSc Advanced Computer Science · Queen Mary University of London · 2025 – Present

Cloud Computing · Machine Learning · Security & Authentication · Functional Programming · Data Analytics · Research Project

Final Year Project: Lightweight Software Verification Tool — designed and implemented a CTL model checker in Python achieving 34/34 test cases passing. Covers Kripke structure parsing, temporal logic evaluation, and automated counterexample generation.

BSc (Hons) Computer Science — 2:1 · Staffordshire University · 2022 – 2025

Full-Stack Development · Cloud Computing · AI · Cybersecurity · Mobile Development · Databases · Networking · UX Design

PROFESSIONAL EXPERIENCE

IT Support Engineer (Deployment / Break-Fix) · Bank of America · via Manpower (Zoom IT) · Apr 2022 – Present

- Maintained enterprise desktop infrastructure in a highly regulated financial environment, ensuring full compliance with FinTech and GDPR security standards.
- Diagnosed and resolved complex hardware and software incidents across multiple business units, consistently minimising downtime.
- Led large-scale device deployment and asset management programmes supporting hundreds of end-users across enterprise rollouts.
- Implemented security patches, system hardening, and OS updates aligned with financial industry regulatory requirements.

Field Support Technician — Level 2 · Amadeus Global Travel Technology · via Manpower (Zoom IT) · 2023 – 2024

- Delivered secure laptop and mobile device replacements with enterprise application installations adhering to strict security and compliance protocols.
- Provided Level 2 escalation support for complex technical incidents, collaborating cross-functionally to reduce mean time to resolution.
- Maintained detailed documentation of system configurations, troubleshooting procedures, and incident records.

IT Technician / Support · ENI Energy · via Manpower (Zoom IT) · 2022 – 2023

- Provided frontline hardware, software, and network connectivity support, maintaining high system availability across the organisation.
- Proactively monitored enterprise uptime and built out the team knowledge base with structured technical documentation.

TECHNICAL SKILLS

Languages: Python, Java, JavaScript (ES6+), TypeScript, C# (.NET), HTML5, CSS3

Frameworks & Libraries: React, Next.js, Node.js, Express.js, Scikit-learn, NumPy, Pandas, Matplotlib

Cloud & DevOps: AWS (EC2, S3, Lambda, IAM, CloudWatch), Docker, Kubernetes, Azure AD, Microsoft Intune, Active Directory, CI/CD, Git, Linux/Unix, Windows Server

Security: AES-256 Encryption, JWT Auth, IAM, CVE Analysis, Network Security, Secure Design Patterns, FinTech/GDPR Compliance

AI & Data: Machine Learning, NLP, Statistical Analysis, Predictive Modelling, Data Visualisation, Model Evaluation

Methodologies: Agile / Scrum, RESTful API Design, Version Control (Git), OOP, Functional Programming, Formal Verification

Databases: SQL, MySQL, PostgreSQL, SQLite, Query Optimisation, ER Modelling

KEY PROJECTS

CTL Model Checker (LSVT) — Python-based Computation Tree Logic model checker with Kripke structure support, 34/34 tests passing, counterexample generation, and benchmark suite. Submitted as MSc research project.

SOC Command Center — Enterprise SOC dashboard with real-time threat detection, live attack visualisation, incident response workflows, and GDPR/SOC2/ISO27001 compliance tracking.

CipherOS — Cryptography Toolkit — Client-side browser toolkit implementing AES-256, SHA/MD5 hashing, JWT decoding, and password analysis via Web Crypto API — zero server-side dependencies.

NetScan Pro — Vulnerability Scanner — Nmap/Nessus-style scanner simulation with 4 scan modes, CVE database of 10,000+ entries (Log4Shell, EternalBlue), and custom scan algorithms.

Enterprise Analytics Dashboard — Next.js analytics platform with real-time KPIs, AI-powered revenue forecasting (94% confidence interval), live user tracking, and interactive Recharts visualisations.